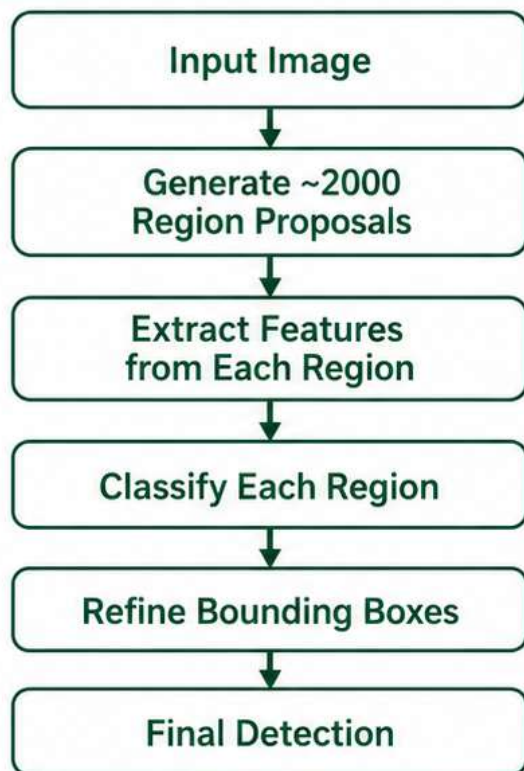


The Problem with Traditional Object Detection



Before YOLO

Multi-Step Detection Pipeline

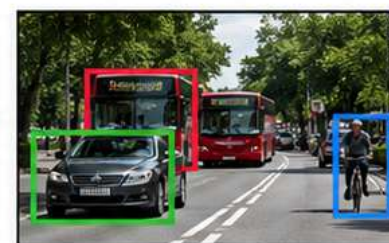
- Generate approximately 2000 candidate regions
- Run feature extraction on every region
- Classify each region independently
- Refine bounding box coordinates
- Combine all results into final detections



Input Image



~2000 Region Proposals



Final Detections



Major Issue

Traditional detectors required multiple expensive stages and could take **45+ seconds per image**, making real-time applications impossible.

Real-World Need for Faster Object Detection



The Challenge

Traditional object detectors were accurate, but far too slow for real-world deployment.

Modern applications require predictions in **milliseconds**, not seconds.



Traditional Detection
45+ Seconds/Image



Not Practical



Need for Real-Time AI

Why Speed Matters in Object Detection

Enter YOLO (2016)



The Key Insight

Instead of searching thousands of regions separately, YOLO predicts all objects in a single forward pass.



Traditional detectors

Analyze many candidate regions independently



YOLO

Looks at the entire image once and predicts all detections together



Traditional Detection
≈45+ seconds per image



YOLO
≈30 milliseconds per image

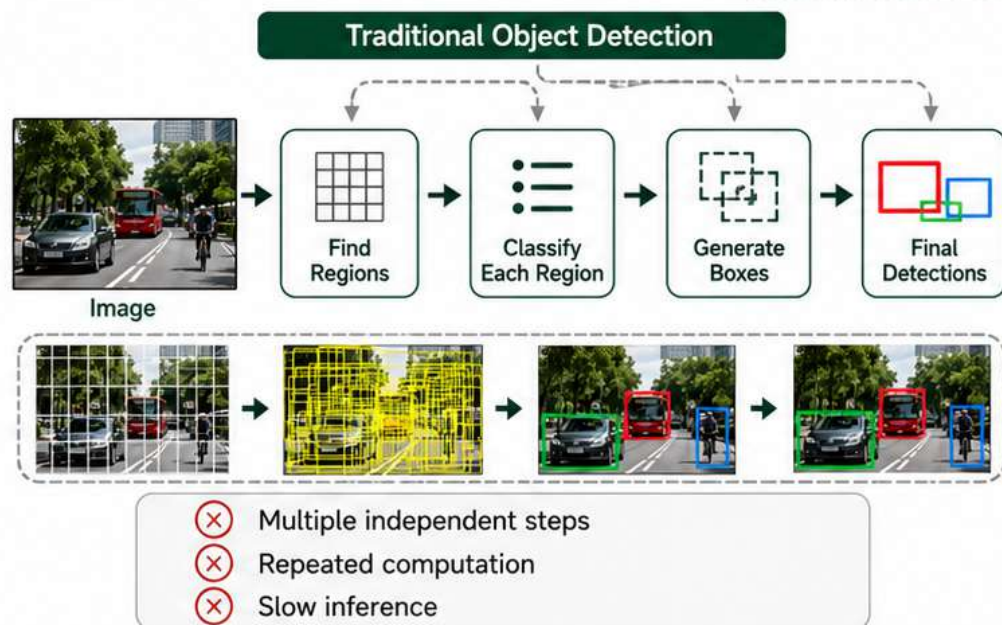


RESULT: ≈45 seconds → ≈30 milliseconds
Massive speed improvement for real-time applications.

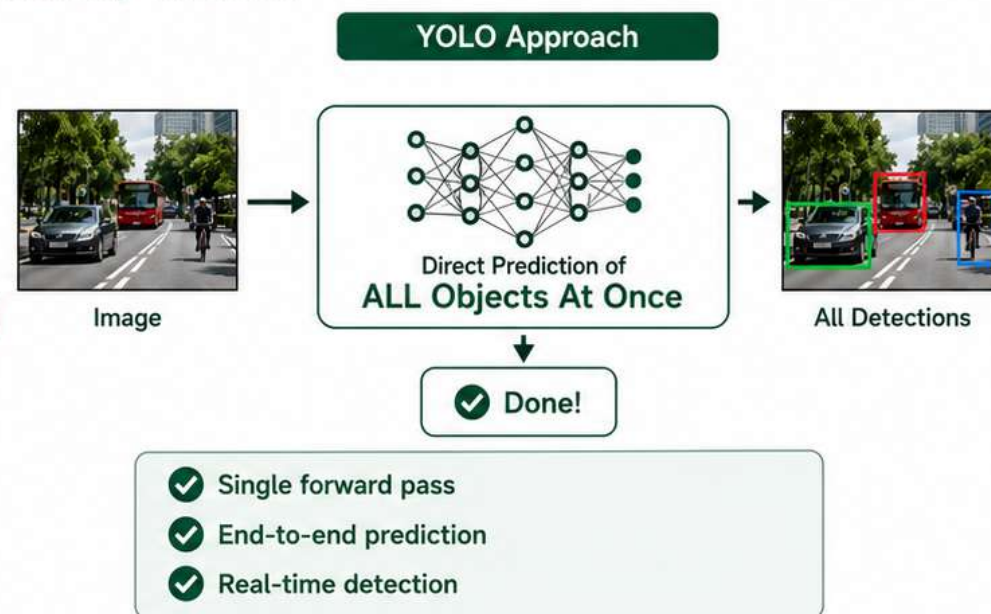
YOLO: You Only Look Once (2016)

The Core Insight

What makes YOLO fundamentally different?

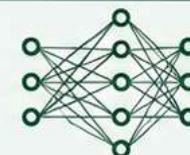


VS



The Breakthrough

Instead of first finding objects and then classifying them, YOLO predicts **all object locations and classes** directly from the image in a **single pass** through the network.



Traditional Detection
Multiple Steps



YOLO
One Neural Network
One Prediction
One Pass

=

ALL
Objects At Once

YOLO Philosophy: You Only Look Once

Grid Division Strategy

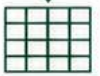
How can one neural network detect many objects at the same time?

Step 1

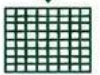
Divide the image into a grid



Image



7 x 7 Grid



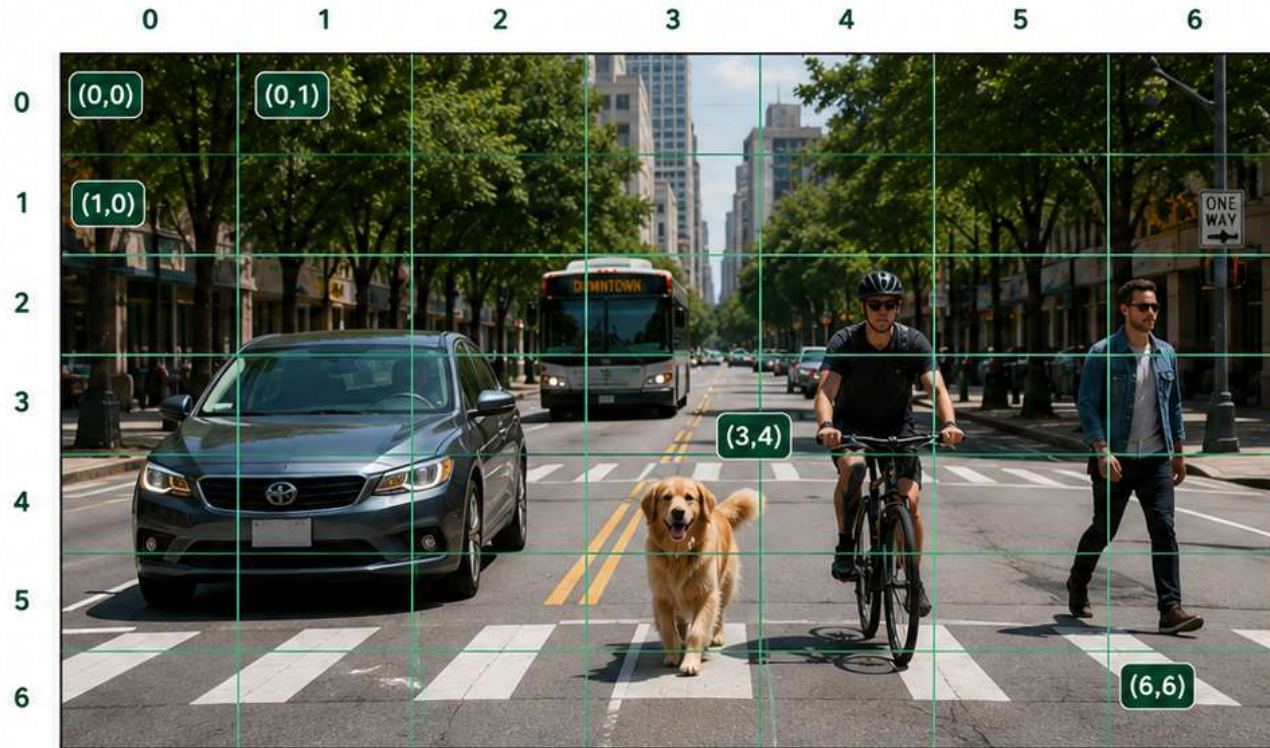
49 Grid Cells

Step 2

Each cell becomes responsible for detecting objects.



One cell = One detector



Key Idea

YOLO divides the image into an $S \times S$ grid.

Example:

- $S = 7$
- $7 \times 7 = 49$ Cells

Each cell independently participates in object detection.



Why Do This?

Instead of searching thousands of candidate regions, YOLO distributes the detection task across grid cells.

Every cell becomes responsible for a small region of the image.



Important Concept



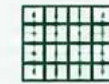
Divide image into $S \times S$ grid (e.g., $7 \times 7 = 49$ cells)



Each grid cell learns to detect objects that belong to its region.



Image



Grid Division ($S \times S$ Grid)

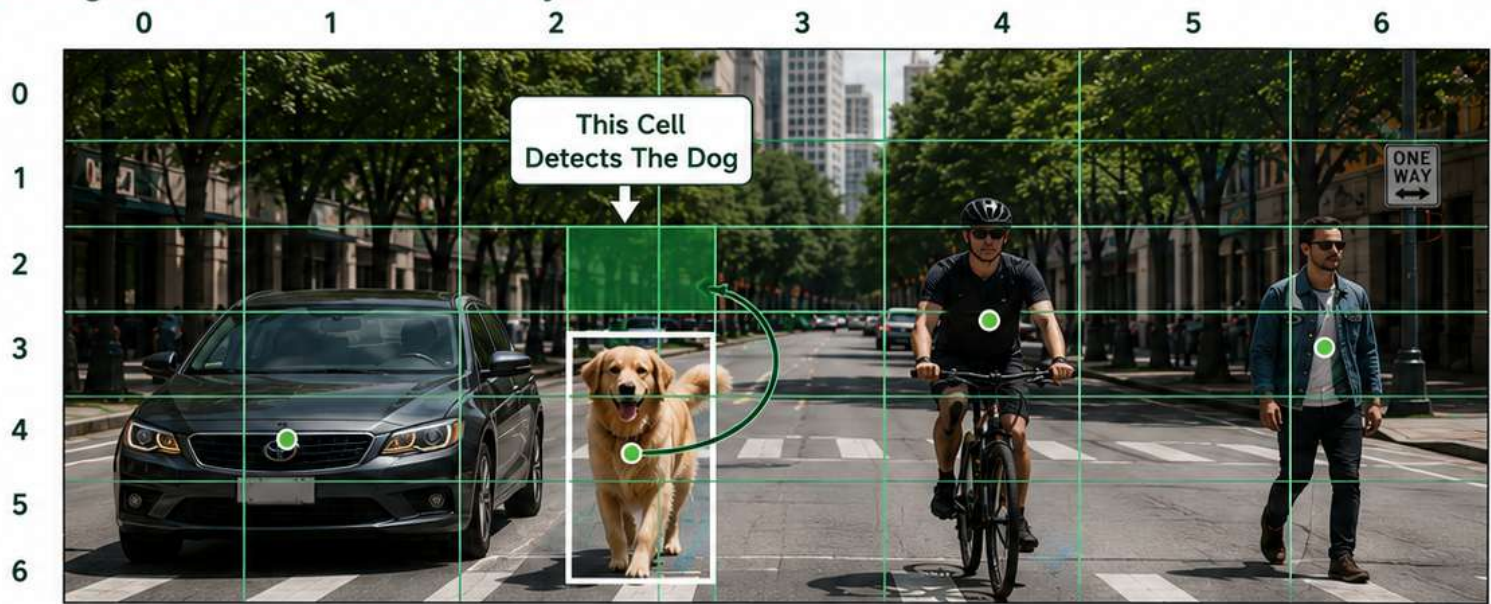


Multiple Detectors Working Together

YOLO Step 1: Divide the Image into a Grid

Object Responsibility

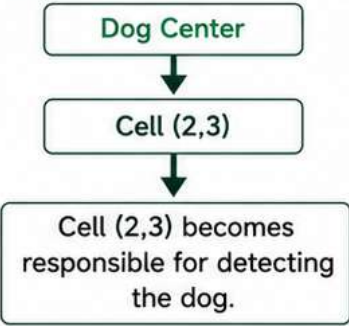
Which grid cell should detect an object?



Key Rule

An object belongs to the cell that contains its center point.

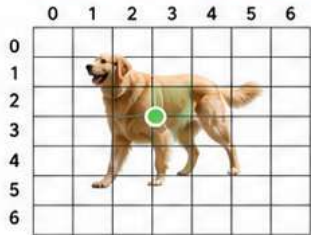
Example



Important Concept

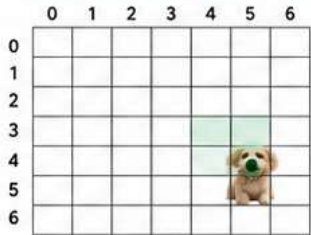
- ✓ Bounding box size does **NOT** matter.
- ✓ Object location does **NOT** matter.
- ✓ Only the **CENTER** determines which cell is responsible.

Example A: Large Dog Spanning Multiple Cells



Center in Cell (2,3)
↓
Responsible Cell = (2,3)

Example B: Small Dog Fully Inside One Cell



Center in Cell (4,5)
↓
Responsible Cell = (4,5)



Object Center



Responsible Grid Cell

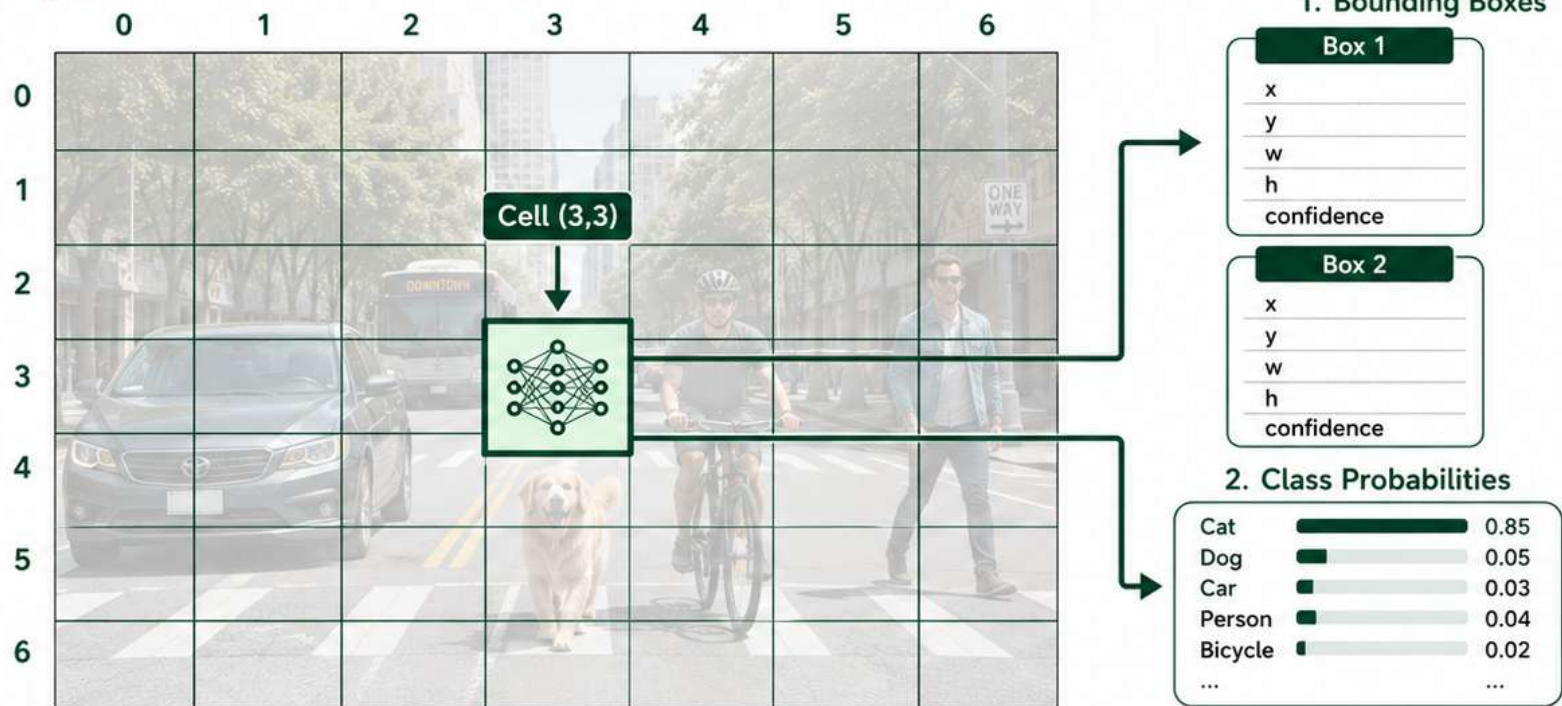


Detection Prediction

YOLO Rule: The Cell Containing the Object Center Owns the Detection

What Does Each Grid Cell Predict?

Every grid cell acts like a mini object detector



Each Cell Predicts

1. Bounding Boxes

Used to locate objects.

Contains:

- x
- y
- w
- h
- confidence

2. Class Probabilities

Used to identify object category.

Examples:

- Dog
- Cat
- Car
- Person
- ...



Key Idea

Every grid cell predicts:



Where an object is



Whether an object exists



What type of object it is



Grid Cell



Bounding Boxes
(Location + Confidence)



Class Probabilities
(Object Category)

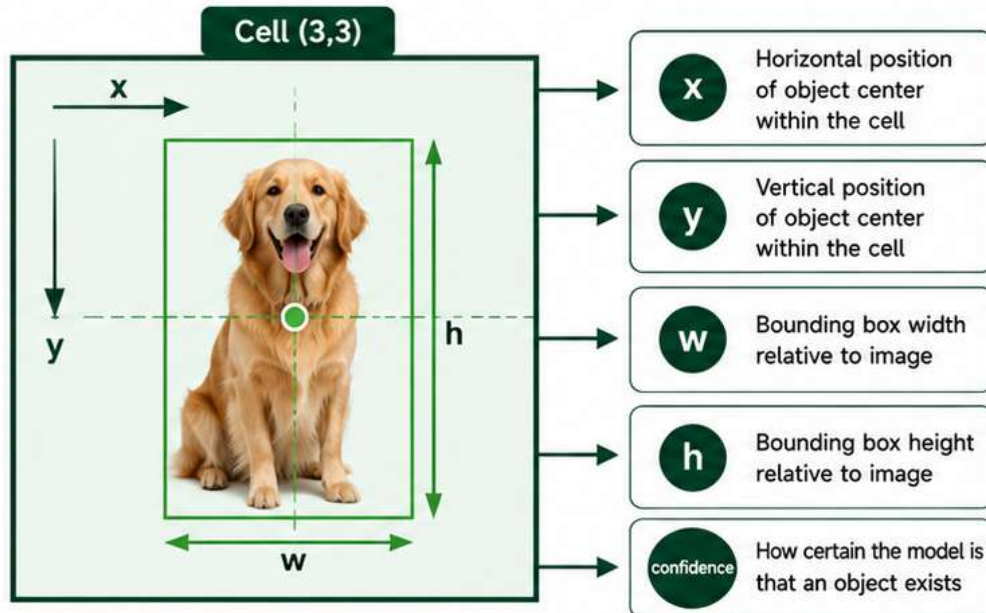


Object Detection
(Final Prediction)

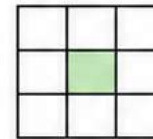
YOLO Step 2: Every Grid Cell Produces Detection Predictions

Bounding Box Predictions

How does a grid cell describe an object?



Grid Cell



Bounding Box Prediction

(x, y, w, h, confidence)

Example Prediction

x = 0.35
y = 0.70
w = 0.40
h = 0.50
confidence = 0.92



Interpretation



Object center lies inside the cell.



Object occupies 40% image width.



Object occupies 50% image height.



Model confidence = 92%.



Key Idea

Each bounding box is represented by

(x, y, w, h, confidence)

These values tell YOLO:

- Where the object is
- How large it is
- How confident the model is



Location
(x, y)

+



Size
(w, h)

+



Certainty
(confidence)



Grid Cell
(3,3)



(x, y, w, h, confidence)
Bounding Box Prediction

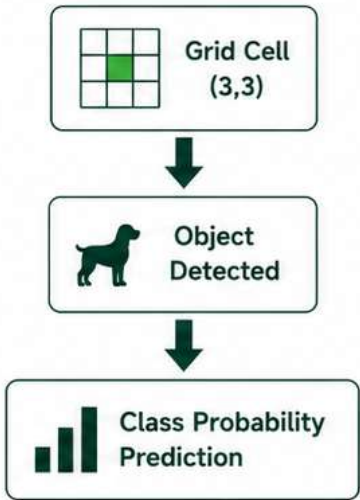
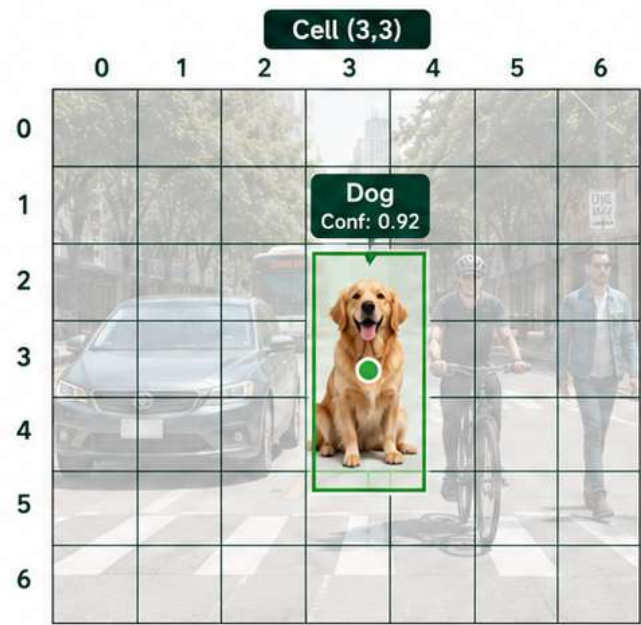


Bounding Box
on the object

YOLO Bounding Box Representation

Class Probabilities

How does YOLO identify what object it has detected?



Predicted Classes	
Class	Probability
Dog	0.92
Cat	0.04
Car	0.02
Bicycle	0.01
Person	0.01

Dog	Cat	Car	Bicycle	Person
0.92	0.04	0.02	0.01	0.01

? What Are Class Probabilities?

YOLO predicts:

- $P(\text{Class}_1)$
- $P(\text{Class}_2)$
- $P(\text{Class}_3)$
- ...
- $P(\text{Class}_n)$

Examples:

- $P(\text{Dog})$
- $P(\text{Cat})$
- $P(\text{Car})$
- $P(\text{Person})$
- ...

Example

$P(\text{Dog}) = 0.92$
 $P(\text{Cat}) = 0.04$
 $P(\text{Car}) = 0.02$
...

Prediction:

DOG

Σ Mathematical Representation

[Dog, Cat, Car, Person, Bicycle]

↓

[0.92, 0.04, 0.02, 0.01, 0.01]



Key Insight

Class probabilities tell YOLO what the detected object is.

Bounding boxes tell YOLO where the object is.

Bounding Box Output

Answers: Where?

Shows: x, y, w, h

Class Probability Output

Answers: What?

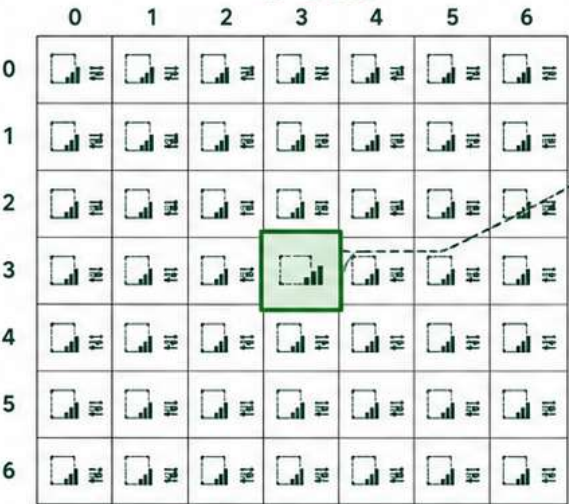
Shows:



Final Output

What does YOLO produce after processing the entire image?

7 × 7 Grid
49 Cells



Zoom: Cell (3,3)

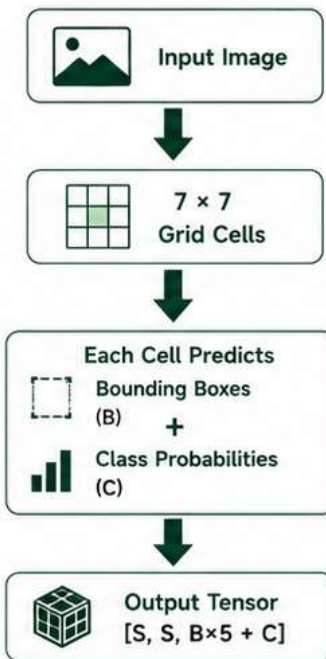
Bounding Box Predictions

Box 1
x, y, w, h, conf.

Box 2
x, y, w, h, conf.

Class Probabilities

80 values
⋮



Output Shape

[S, S, B × 5 + C]

Example

S = 7
B = 2
C = 80

↓

[7, 7, 2 × 5 + 80]

↓

[7, 7, 90]

90 Values Per Cell

What Is Inside One Cell?

Box 1: x, y, w, h, confidence

Box 2: x, y, w, h, confidence

Class Probabilities: 80 Values

Total

10 Bounding Box Values
(2 boxes × 5 values)
+
80 Class Values
=
90 Outputs

90 Outputs Per Cell

Key Idea

YOLO predicts everything at once. Every grid cell contributes its own detection predictions, creating one large output tensor.

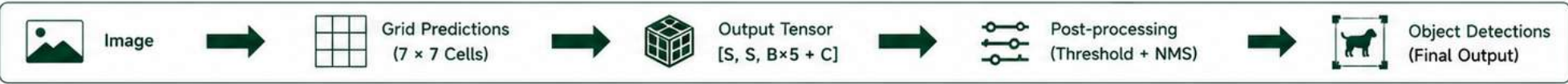
Visualizing the Output Tensor

7 Grid Rows (S)
7 Grid Columns (S)
90 Prediction Values (B×5 + C)

Each cell generates 90 prediction values that become part of the final output tensor.

What Happens Next?

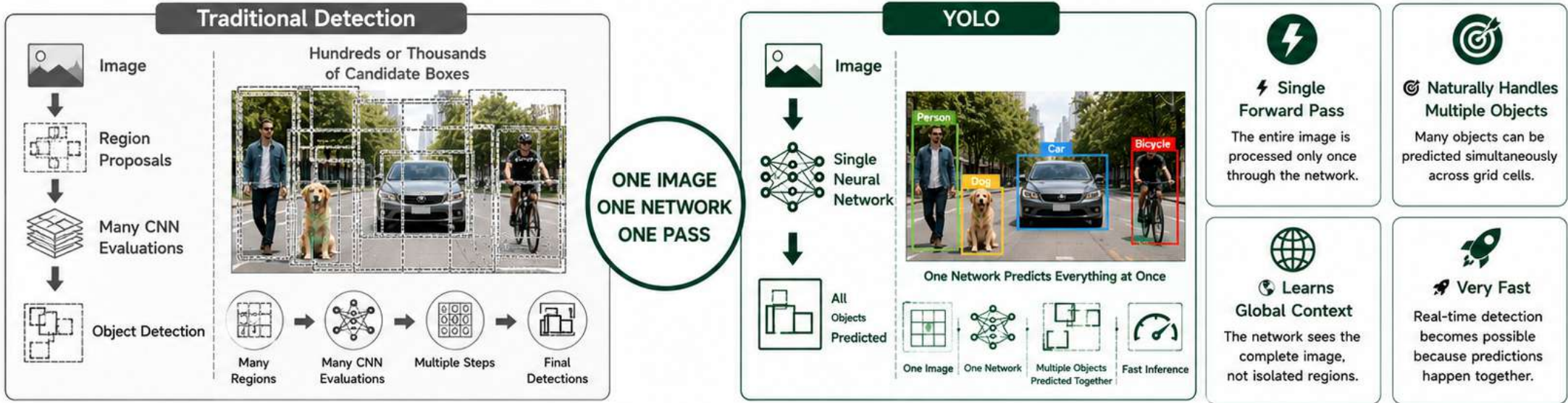
```
graph LR; A[Use predictions from all cells] --> B[Apply confidence thresholding + NMS] --> C[Final Object Detections];
```



YOLO Output Representation: [S, S, B×5 + C]

Why This Works

Why is YOLO so fast and effective?



Confidence Score Definition

How confident is YOLO that an object exists?

P(Object)

Object Present

 $P(\text{Object}) = 1$ 

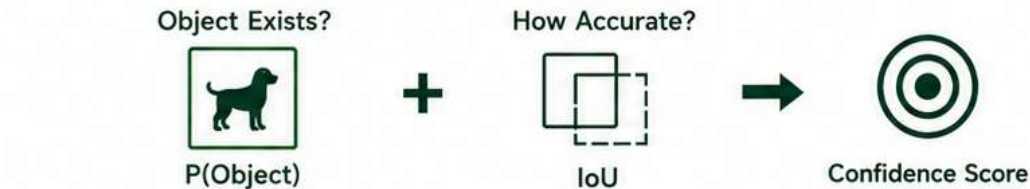
No Object Present

 $P(\text{Object}) = 0$ 

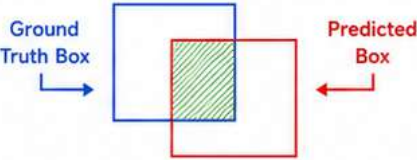
Confidence

=

$P(\text{Object}) \times \text{IoU}(\text{Predicted Box, Ground Truth Box})$



IoU (Intersection over Union)



$\text{IoU} = \frac{\text{Intersection}}{\text{Union}}$

Example 1: Perfect Detection

 $P(\text{Object}) = 1$
 $\text{IoU} = 1.0$
↓
Confidence = 1.0

Example 2: Partial Detection

 $P(\text{Object}) = 1$
 $\text{IoU} = 0.6$
↓
Confidence = 0.6

Example 3: No Object

 $P(\text{Object}) = 0$
 $\text{IoU} = 0$
↓
Confidence = 0.0

Real Example: Dog Detection

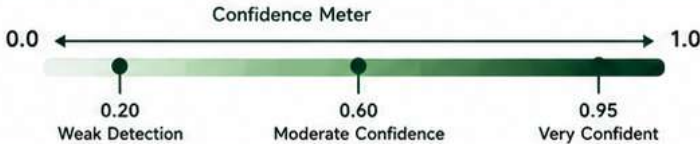
 $\text{IoU} = 0.72$
 $P(\text{Object}) = 1$
Confidence = 0.72



Key Insight

Confidence answers two questions:

- 1 Is there an object?
- 2 How well does the predicted box match the actual object?



Object Presence
(Is there an object?)

×



Localization Quality
(How accurate is the box?)

=

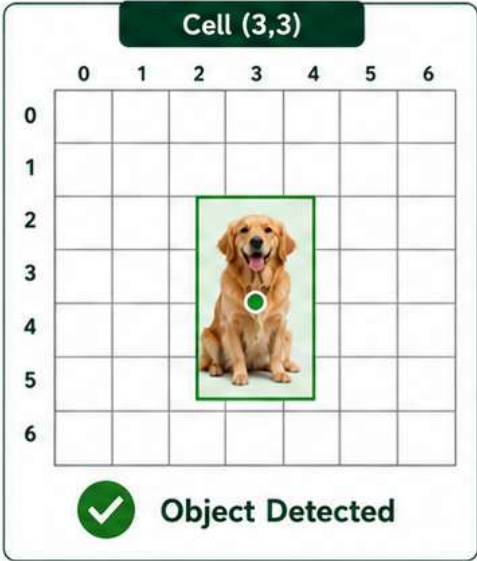


Confidence Score
(How confident is the detection?)

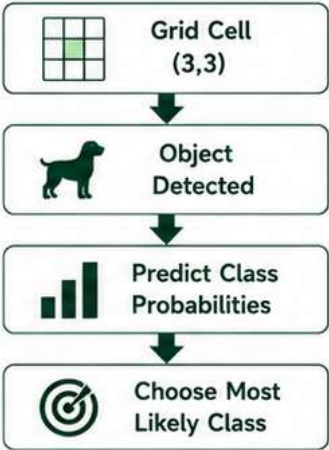
Confidence = $P(\text{Object}) \times \text{IoU}$

Class Probability

What type of object did the grid cell detect?



$P(\text{Class}_i \mid \text{Object Detected In Cell})$



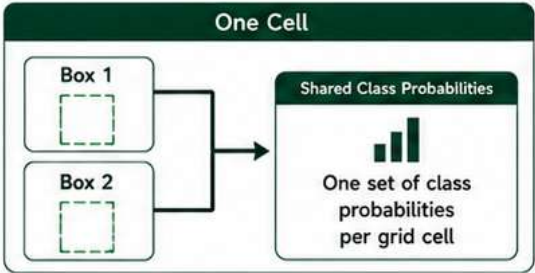
Predicted Classes		
Class		Probability
Dog		0.92
Cat		0.04
Car		0.02
Person		0.01
Bicycle		0.01

Example Prediction

$P(\text{Dog}) = 0.92$
 $P(\text{Cat}) = 0.04$
 $P(\text{Car}) = 0.02$
 $P(\text{Person}) = 0.01$
 $P(\text{Bicycle}) = 0.01$

↓

Prediction:
DOG



Important

YOLO predicts class probabilities for the entire grid cell.

The probabilities are shared across all bounding boxes in that cell.

Mathematical Interpretation

$P(\text{Dog} \mid \text{Object})$
 $P(\text{Cat} \mid \text{Object})$
 $P(\text{Car} \mid \text{Object})$
 $P(\text{Person} \mid \text{Object})$
...

Conditional Class Probabilities

Visual Example (Cell 3,3 Detecting a Dog)

Dog 92%
Cat 4%
Car 2%
Person 1%
Bicycle 1%

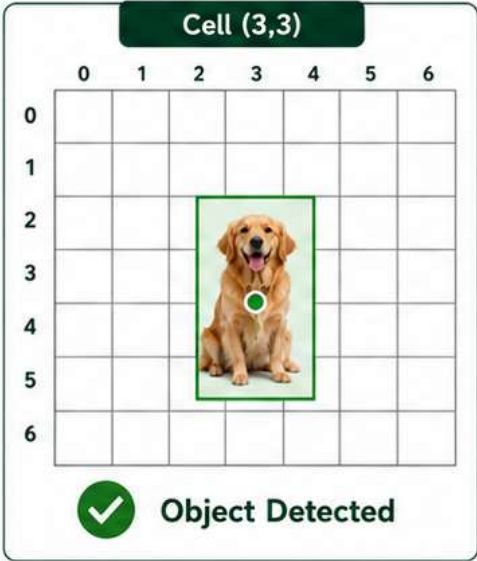


Bounding Boxes tell YOLO **WHERE** the object is.

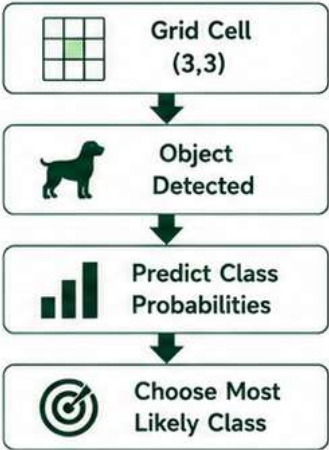
Class Probabilities tell YOLO **WHAT** the object is.

Class Probability

What type of object did the grid cell detect?



$P(\text{Class}_i \mid \text{Object Detected In Cell})$



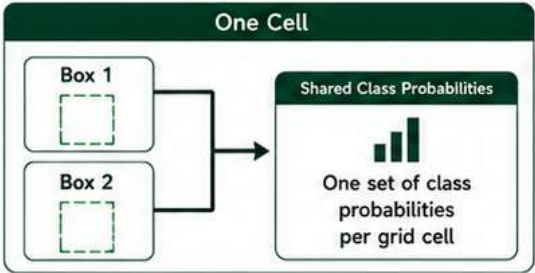
Predicted Classes		
Class		Probability
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Cat		0.04
Car		0.02
Person		0.01
Bicycle		0.01

Example Prediction

$P(\text{Dog}) = 0.92$
 $P(\text{Cat}) = 0.04$
 $P(\text{Car}) = 0.02$
 $P(\text{Person}) = 0.01$
 $P(\text{Bicycle}) = 0.01$

↓

Prediction:
DOG



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$P(\text{Dog} \mid \text{Object})$
 $P(\text{Cat} \mid \text{Object})$
 $P(\text{Car} \mid \text{Object})$
 $P(\text{Person} \mid \text{Object})$
...

Conditional Class Probabilities

Visual Example (Cell 3,3 Detecting a Dog)

Dog 92%
Cat 4%
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Person 1%
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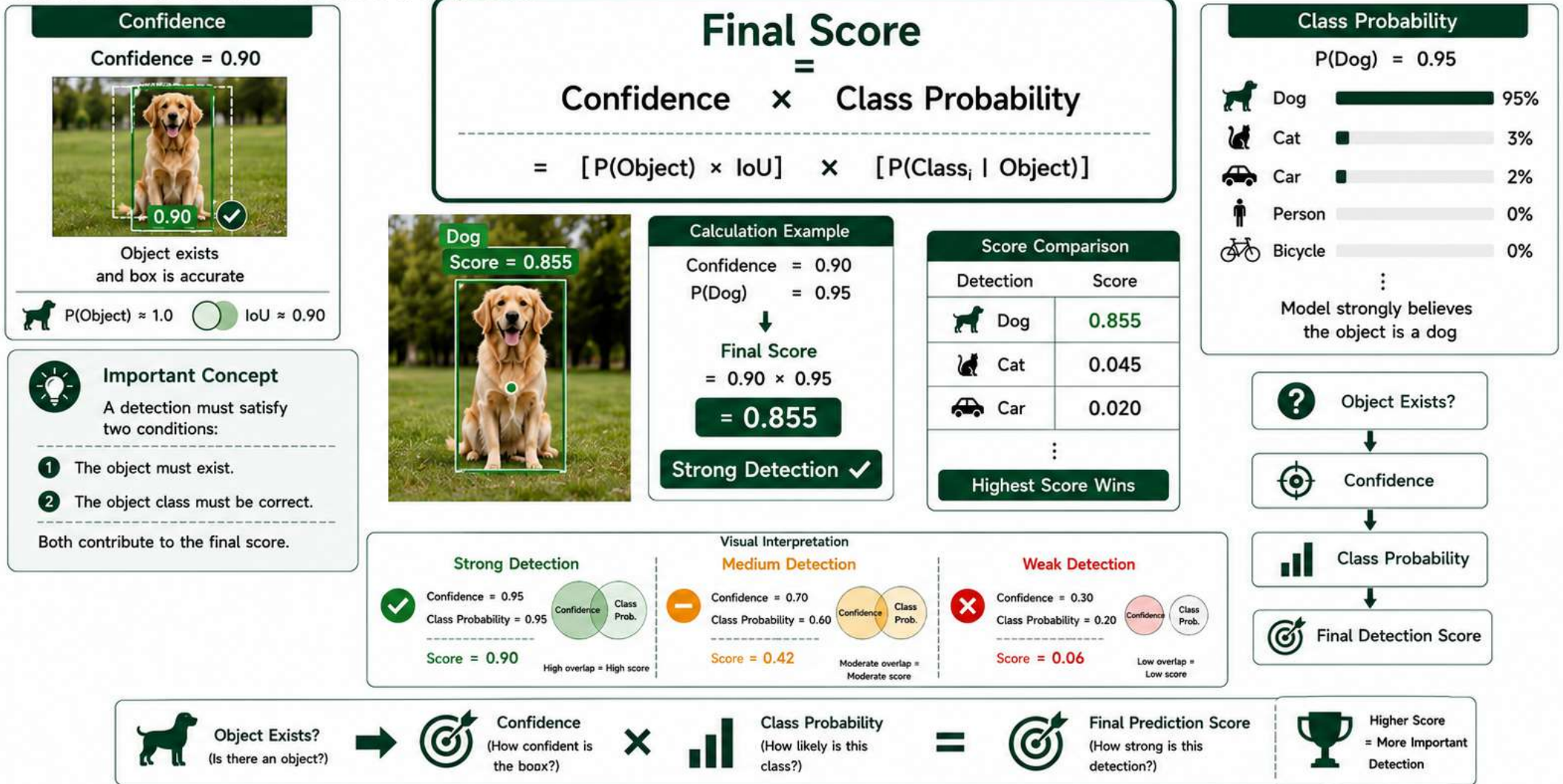


Bounding Boxes tell YOLO **WHERE** the object is.

Class Probabilities tell YOLO **WHAT** the object is.

Final Prediction Score

How does YOLO decide which detections are important?



YOLO Detection Score Calculation